Ultrafiltration

Pre-Lab Plan

Unit Operations Lab 1

1/22/19

Group 3, Tuesday AM

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1. Experimental Objective
   The objective in this experiment is to determine the permeability of the hollow fiber membrane with a water and water-milk solution, calculating the filtering effectiveness, and the concentration factor. This will be done by determining the flux of both liquids at several different outlet pressures and percent recoveries.
2. Experimental
   A. Apparatus

The apparatus will be broken into three sections: the first being the front, the second being the left side of the back, and the last being the right side of the back.

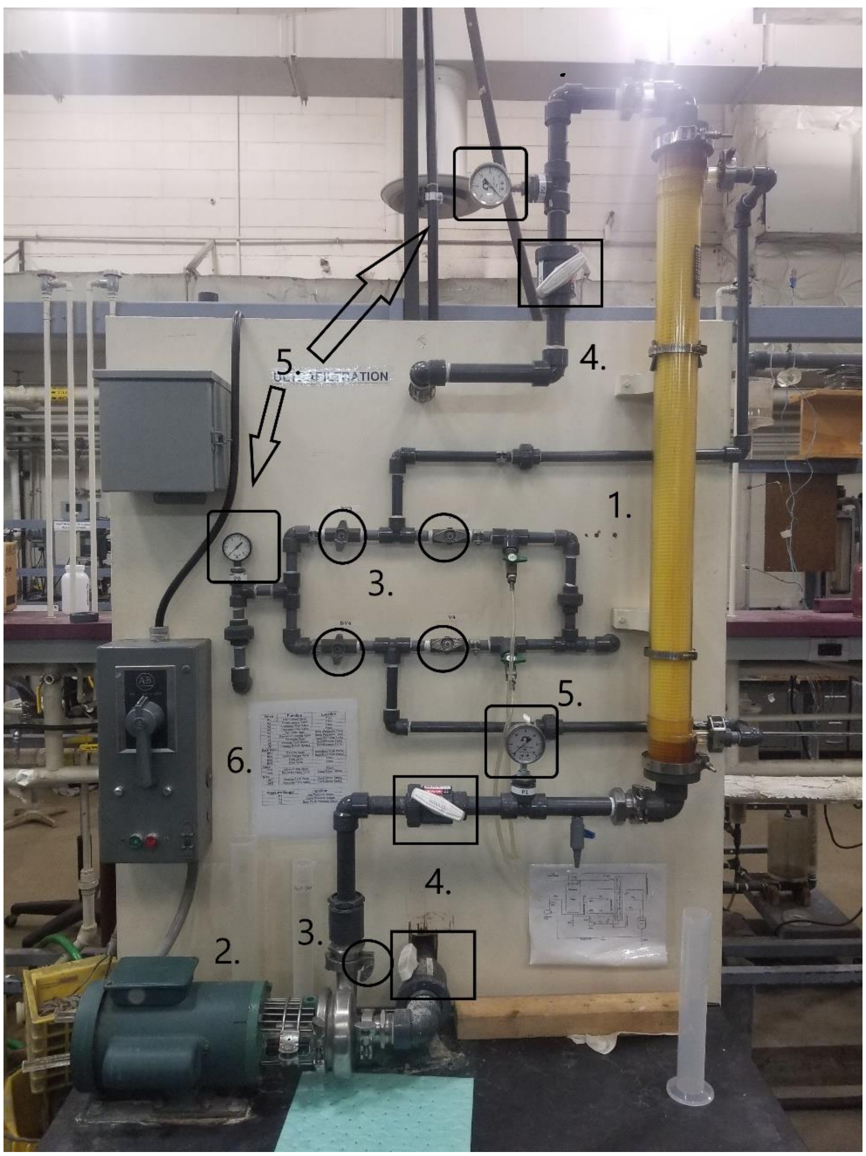

Figure 1: Main Apparatus with Labeled Components

Figure 2: Left Side View of Storage Tank

Figure 3: Right Side View of Storage Tank

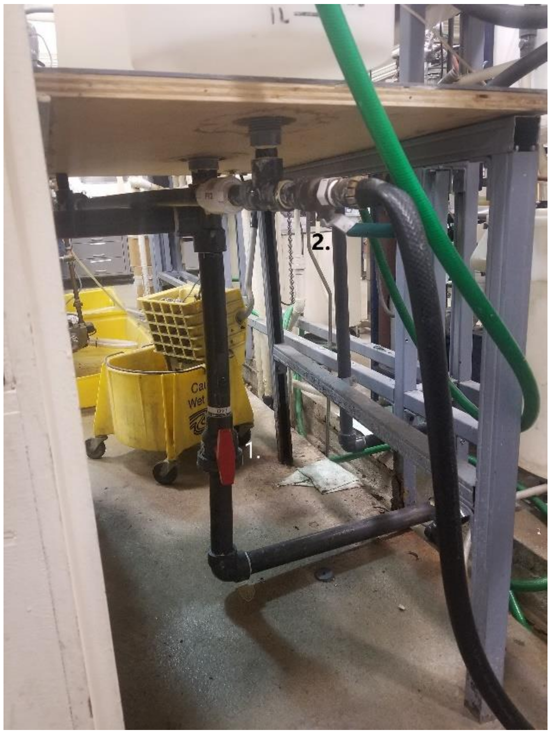

Figure 4: Drainage Area of Storage Tank and System

Figure 5: Temperature Reading and Deionized Water Feed Line

Table 1: Description of Key Components and how they are Used in Figure 1

	Component	Description
1.	Hollow Fiber Membrane	Separates the milk proteins from the solution.
2.	Circulation Pump	Pumps the solution to the system.
3.	Flow Control Valves	Control the flows (all circled).
4.	Pressure Valves	White valves that control pressure (boxed).
5.	Pressure Gauges	Reads the pressure in psi (rounded boxes).
6.	Power Supply	Supplies power to the system.

Table 2: Description of Key Components and how they are Used in Figure 2

	Component	Description of Component
1.	Process Tank	The milk solution is stored in the tank before being pumped into the system.
2.	Back-Flush Pump	Used in the cleaning of the system
	Flow Control Valves	Control the Flow (Valves circled)

Table 3: Description of Key Components and how they are Used in Figure 3

	Component	Description of Component
1.	Permeate Tank	The various solutions are stored in the tank before being pumped into the system.
	Flow Control Valves	Control the flow (valves circled).

Table 4: Description of Key Components and how they are Used in Figure 4

	Component	Description of Component
1.	Drainage valve for the system	Drains the filtration part of the system.
2.	Drainage valve for the process and permeate tanks.	Drains the tanks upon completion.

Table 5: Description of Key Components and how they are Used in Figure 2

	Component	Description of Component
1.	Analog Meter	Reads the temperature of the DI water coming into the system.
2.	DI Needle Valve	Controls the flow of the DI water to the steam heat exchanger.
3.	Flow Control Valve	Controls the flow of the heated DI water to the process tank.

B. Materials and Supplies

A table of the materials and supplies used in the experiment are provided in Table 6.

Table 6: Materials and Supplies

	Material/Supply	Description
1.	Milk	Will be mixed with DI water
2.	Deionized water	Used to clean and make the solution
3.	Phosphoric Acid	Used to clean the system
4.	Sodium Hydroxide	Used to clean the system
5.	pH probe	Used to measure the pH of acid solution
6.	Scale	Used to measure chemicals for the various solutions
7.	Beaker	Used in the making of the solutions
8.	Stopwatch	Used to measure time intervals
9.	Goggles	Used for safety
10.	Gloves	Used for safety
11.	Mop and Bucket	Used to for spills
12.	Graduated Cylinder	Used to measure collected permeate

C. Experimental Plan

1. Cleaning Procedure
    a. Liquid Solution Flush
        i. Fill the process tank with 20L of DI water that is around 105°F. Open Intake-Pump Valve (PV1).
        ii. Direct the permeate outlet flow to the process tank.
        iii. Close V1 and open V2.
        iv. Start the circulation pump.
        v. Slowly open V1 until a pressure of 20 psig is read on P1.
        vi. Slowly close V2 until a pressure between 8-10 psig is read on P2.
        vii. Readjust V1 until P1 reads 25 psig and P2 reads 10 psig.
        viii. Allow the system to run for 10-15 minutes. *Check for leaks and make sure there are no air bubbles present.
        ix. Slowly open V2 until it is all the way open.
        x. Close V1.
        xi. Stop the circulation pump.
        xii. Drain the water from the process tank and filter.
        xiii. Rinse the tank.
    b. Acid Cycle
        i. Make 20L of a 2.05 pH phosphoric acid solution in DI water at 105°F.
        ii. Repeat steps ii-vii from Liquid Solution Flush.
        iii. Run for 10 minutes. Then close V3 and V4 and allow the acid solution to run through the filter for another 5 minutes.
        iv. Repeat steps ix-xiii from Liquid Solution Flush.
    c. Drain and Rinse Cycle
        i. Refill the process tank with 50L of 105°F DI water.
        ii. Direct the permeate outlet flow to the permeate tank.
        iii. Open DV2.
        iv. Repeat steps iii-vii from the Liquid Solution Flush. Momentarily crack the drain valves during the rinse cycle to drain out any residual acid. *Ensure the water level of the process tank does not fall below the level of the pump inlet.
        v. Rinse permeate tank.
    d. Caustic Cycle (Save Permeate)
        i. Make up 20L of 1% sodium hydroxide solution in 105°F DI water. Use the weight basis of 8.33lbs water per gallon
        ii. Repeat steps ii-xiii of the Liquid Solution Flush except direct the permeate to the permeate tank and not the process tank. It will be saved for back flushing.

e. Back Flush Mode
    i. For the permeate area, close V6 and open PV2. Close DV2 and open V5. In the back, close BV1 and open BV2. Now open BV4, BV3, and V2. All other valves should be close except for PV1.
    ii. Turn on back flush pump.
    iii. Allow the permeate liquid to flow from the permeate tank, through the system via back flush and into the process tank.
    iv. Once the permeate tank level is half the original level, close DV1.
    v. Drain the permeate tank, close all back flush valves and open the outlet pressure valve.
    vi. Turn off the backflush pump and turn on the circulation pump.
    vii. Slowly open V1 until it reaches 20 psig on P1.
    viii. Slowly close V2 until it reaches 10 psig on P2.
    ix. Adjust the previous two valves until P1 reads 25 psig.
    x. Run the system for 15 minutes.
    xi. Check for any leaks or air bubbles.
    xii. Slowly open V2 and close V1.
    xiii. Drain the filter and the process tank.
    xiv. Rinse the process tank to prepare for the next cycle.

f. Sanitation Cycle
    i. Make a 200ppm sodium hypochlorite solution in 20L of DI water that is at 105°F. (Use 350 mL of bleach/50 L DI water).
    ii. Repeat steps ii-xiii of the Liquid Solution Flush and then repeat the Drain and Rinse Cycle.

2. Experimental Procedure
    g. Water Flux Test
        i. Set the inlet pressure of the system to 25 psig.
        ii. Determine the water flux at outlet pressures of 5, 10, 20, and 15 psig using 20L of 105° DI water. *Ensure the permeate is collected in a suitable graduated cylinder to allow 30 seconds of recovery.
    h. Skim-Milk Test
        i. Fill the process tank with 80L of 105°F DI water and add 1 gallon of skim milk and stir. *Ensure the feed material is at the proper temperature of 105°F.
        ii. Bring the inlet pressure to 25 psig and the outlet to 5 psig. Run at these conditions until 8.379 liters are removed (10% recovery).
        iii. Now return the permeate line back to the process tank to maintain concentration. Determine the permeate flux during this time and be sure to pour the collected amount back to the process tank.

iv. Repeat the procedure with 25% and 40 % recoveries.
v. Vary the outlet pressure in this order: 5, 15, 10, 20, and 5 psig. Allow two minutes after each pressure change to collect the data.

Table 6: Independent and Dependent Variables

Independent Variables	**Dependent Variables**	**Measuring Dependents**
Pressure	Flux	Calculated by the permeate flowrate and the area of the membrane.
Inlet Flowrate	Permeability	Measured by the flux and the applied pressure.

Project Safety Evaluation (10 pts)

Project Title: Ultrafiltration

Date: January 22, 2019

	Yes/No	
Potential electrical hazards?	YES	If yes, identify conditions: The use of a pump is used to feed the membrane filter, water onto the electrical components can short out the pump and cause bodily harm.
Potential mechanical hazards?	YES	If yes, identify conditions: Many valves are used throughout the unit, make sure they are used properly, and the configurations are safe throughout the experiment.
Potential pressure hazards?	YES	If yes, identify conditions: Operating at inlet pressures of 25 psig, and a maximum of 15 psig outlet pressure. Anything higher may damage the membrane in the filter, as well as the pipes throughout the system.
Potential temperature hazards?	YES	If yes, identify conditions: Water used in this experiment is at 105 F and is hot to the touch may cause minor burning if touched unexpectedly. Steam is used to heat the DI water.
Hazardous/toxic chemicals involved?	YES	If yes, hazards involved: Working with strong acids and bases, as well as chlorine, and bleach. These chemicals can cause severe burns if touched on skin, and irreversible damage to eyes if splashed onto face.
Gloves required?	YES	If yes, specify type: Chemical resistant and heat resistant gloves.
DS reviewed by all team members?	YES	List MSDS sheets reviewed: Water, Phosphoric Acid 85+%, Sodium Hydroxide, Sodium Hypochlorite.
Process Parameters	*Max* *Expected*	*List location and possible hazards and precautions*

Temperature (°F)	105 °F	
Pressure (psig)	25 psig	
Safety Controls	*yes/no*	*If yes, Provide description*
	YES	Two valves (V1 & V2) that can be adjusted to control the inlet and outlet pressure.

I have read relevant background material for the Unit Operations Laboratory entitled: "Ultrafiltration" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

1/22/19

signature of team member date

1/22/19

signature of team member date

Matt Wagner 1/22/19

signature of team member date

1/22/19

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1/22/19

Signature of team member date